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Cellulose nanocrystals extracted from rice husk using the formic/peroxyformic acid process: isolation and structural characterization†

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Cellulose derived from biomass is a renewable resource with numerous applications. Using formic/peroxyformic acid at atmospheric pressure, cellulose nanocrystals (CNC) were isolated from rice husk (RH) in this study. This method was an excellent way to get rid of lignin and hemicelluloses from RH. The cellulose was subsequently acid hydrolyzed by H₂SO₄ (64%) for 30 minutes at 45 °C. The chemical and microstructure analysis showed that the lignin and hemicellulose contents of raw RH had been eliminated, and the crystallinity content of CNC was 67.16%. According to transmission electron microscopy (TEM) morphological analysis, CNC measured 19 ± 3.3 nm in diameter, 195 ± 24 nm in length, and 10.2 ± 6.8 in aspect ratio. The thermal stability of RH and CNC was also investigated using thermogravimetric analysis (TGA). These encouraging findings demonstrated the potential for reusing RH agricultural waste to create CNC and include nanocomposites as a reinforcing material.

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1. Introduction

This century's "bioeconomy" paradigm promotes using renewable materials over non-renewables for economic success and environmental sustainability.¹ With an annual production of approximately 7.50 × 10¹⁰ tons, cellulose biopolymer is one of the most abundant materials. Cellulose comprises repeated D-glucopyranosyl units connected by β-(1-4)-glycoside bonds.² Due to the interaction of high molecular weight polymer chains, both crystalline and amorphous regions are presented in cellulose. Crystalline particles called cellulose nanocrystals (CNC) are created when amorphous cellulose is hydrolyzed with acid.³ CNC has several notable characteristics, including its low cost, lack of toxicity, exceptional thermal stability, optical transparency, and biodegradability.⁴ CNC can be extracted from various biomasses and widely used as a polymer composite reinforcement because of its excellent thermal and mechanical properties, including high elasticity modulus (about 150 GPa),⁵ ultralightweight nature (1.6 g cm⁻³),⁶ and large specific surface area (hundreds of m² g⁻¹).⁷ CNC improves several

biodegradable polymers' physicochemical, thermal, and insulating characteristics, making them acceptable for many applications.⁸ Furthermore, CNC can be utilized in the production of barrier films,⁹ shape-memory polymers,¹⁰ bio-nanocomposites,¹¹ drug-delivery materials,¹² photonic crystals,¹³ biomedical devices,¹³ filaments,¹⁴ aerogels,¹⁵ hydrogels,¹⁶ fuel cells,¹⁷ and three-dimensional (3D) printing,¹⁸ as well as for wastewater treatment,¹⁹ agricultural product production,²⁰ adsorption,²¹ and cultural heritage materials.²² The surface chemistry of cellulose derivatives can be altered for numerous additional applications.²³

Current research has focused on extracting CNC from agricultural byproducts using chemical, mechanical, and enzymatic approaches.^{24,25} Due to the inherent drawbacks associated with mechanical techniques, such as their time-consuming nature and the need for high pressure and kinetic energy, as well as the limitations of enzymatic methods, which are often tedious and reliant on the capabilities of microorganisms, chemical methods have emerged as the favored choice for achieving more efficient cellulose isolation. These chemical methods can be employed independently or with mechanical or enzymatic approaches. Several studies have used diverse raw material sources, including sugarcane bagasse,²⁶ coconut fiber,²⁷ rice straw,²⁸ banana plant pseudostems,²⁹ and tea leaf waste.³⁰ The rice husk (RH) is a natural fiber that develops around the rice kernel and is removed during rice production. Prior research focused on the high-temperature treatment of RH to extract silica³¹ or the surface treatment of RH for application as a reinforcing component in lightweight composites.³²

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For the past two decades, in Vietnam, the surface of RH has been treated using various chemical methods as a heavy metal adsorption material or a filler in the fabrication of composite materials.³³ The most recent is using RH as a raw material source for thermally treating silica or activated charcoal. Vietnam is an agricultural country whose principal food source is rice. In 2016, Vietnam's rice production topped 42 million tons, with rice residue accounting for 20% of total production. Most RHs are torched or discarded. When RHs are burned, an enormous amount of ash is formed during carbon metabolism. It exists as thin, light granules that might aggravate human respiratory disorders.³² Furthermore, bacteria decomposing the debris will generate methane gas, which hurts the environment. These variables make processing this waste more difficult. Therefore, more research is needed on utilizing this abundant raw material source.

RH has four essential parts that make up its structure: cellulose (25–35%), hemicellulose (18–21%), lignin (18–21%), and silica (15–17%). The rest comprises soluble chemicals and water (7–15%).³⁴ As a result, research incorporating rice chaff as a raw material source for synthesizing CNC is in high demand. This research aims to isolate cellulose from Vietnamese RH and hydrolyze CNC for reinforced-phase applications in composite materials. The isolation of cellulose was achieved *via* the formic/peroxyformic acid method, comprising three principal stages: formic acid treatment (HCOOH), peroxyformic acid treatment (PFA, a solution of HCOOH and H₂O₂), and bleaching with a solution of NaOH and H₂O₂. As far as the authors know, this is the first study to describe the delignification of lignocellulosic biomass and the extraction of cellulose fibers from RH using a pre-treated process involving formic/peroxyformic acid. For decontaminating biomass, sodium chlorite (NaClO₂), sodium hypochlorite (NaClO), chlorine dioxide (ClO₂), chlorine (Cl₂), and hydrogen peroxide (H₂O₂) were utilized.^{35–37} Although less effective than alternative bleaching agents, H₂O₂ is still regarded as a greener alternative to chlorinated solutions.³⁸ This investigation's unconventional aspect is utilizing an oxidant (H₂O₂) and alkali (NaOH) solution to extract cellulose fibers, depolymerize lignin and remove hemicellulose. Under specific conditions, NaOH, a potent alkaline chemical, can eliminate hemicellulose and lignin.³⁹ Bleaching with an alkaline hydrogen peroxide solution (pH 8–12) aims to enhance the whiteness of the cellulose by dissolving any remaining lignin and hemicellulose. The hydroperoxide anion (HOO[−]) in the alkaline media oxidized the lignin structure, forming more cellulose-rich fibers and a lighter color.⁴⁰

Furthermore, the literature contains several noteworthy reports concerning biomass pre-treatment, which encompass acidic and alkaline treatment, pulping, and bleaching.^{41,42} Implementing acidic or alkaline pre-treatment methods during cellulose isolation increases costs for corrosion-resistant materials and post-processing technologies required to neutralize or eliminate corrosive chemicals. As most inorganic acids are highly corrosive and difficult to recover, conventional inorganic acid pre-treatment violates the rules of sustainable and green chemistry.⁴³ Using vacuum distillation or recrystallization, organic acids can be easily recycled and are less

corrosive than inorganic acids.^{24,44} Another innovation of this study is incorporating formic acid, the simplest carboxylic acid characterized by its low boiling point (100.8 °C)⁴⁵ and relatively strong acidity. This acid was utilized to dissolve hemicellulose partially, cause it to swell, and facilitate lignin migration to the RH surface. As a result, the bleaching stage for cellulose isolation was significantly enhanced in efficiency.

2. Experimental

The RHs originated in the Cu Chi rice-growing region of Ho Chi Minh City. After collection, they were rinsed with water, sun-dried, and ground into a fine powder. Sodium hydroxide (NaOH, 96%), hydrogen peroxide (H₂O₂, 30%), formic acid (HCOOH, 90%), and sulfuric acid (H₂SO₄, 98%) originate from Xilong Scientific Co. China. All are commercial chemicals and were used directly.

2.1. Preparation of cellulose

The sequential process of cellulose isolation from RH involved three primary steps: HCOOH acid treatment, peroxyformic acid (PFA) treatment, and bleaching using a NaOH/H₂O₂ mixture. Initially, the RH powder was immersed in distilled water, followed by boiling at 100 °C for 2 hours to eliminate impurities. Subsequently, the mixture underwent filtration and desiccation at 70 °C for 8 hours. The homogenous blending of RH powder was treated with a 90% HCOOH solution at a ratio of 1 : 15 (RH powder to 90% HCOOH acid volume) at 80 °C. After 2 hours, the sludge was filtered, rinsed with hot water, and desiccated at 80 °C for 6 hours before PFA treatment. A solution of PFA, comprising 90% HCOOH, 4% H₂O₂, and 6% H₂O, was added to RH powder. For 2 hours, the PFA treatment was kept at 80 °C. The filtration method was used to separate the solid product. The substance was carefully washed with distilled water until the pH reached 6–7. After drying for 12 hours at 80 °C, the solid product was bleached with a NaOH/H₂O₂ mixture. Using a 1 M NaOH solution, the pH of the suspension was carefully adjusted to 11. After that, 30% H₂O₂, equal to 40% of the solid mass in the rest, was slowly added. The mixture was then mixed for an hour at 80 °C.

2.2. Production of cellulose nanocrystals (CNC)

To make CNC, RH cellulose was broken down with a 64% sulfuric acid solution (1 : 20 ratio of solid mass to acid volume) at 45 °C for 30 minutes. The mixture was carefully put into a container with 1000 mL of distilled water to finish the process. The solution was diluted with water until pH 7 was obtained. The resultant mixture was centrifuged at 4000 rpm for 10 minutes, twice with distilled water and thrice with ethanol. After drying out at 80 °C until the mass stayed the same, a sample of white powder was taken as the result. CNC yield was determined using RH starting weight.⁴⁶

2.3. Characterizations

2.3.1. Fourier transform infrared spectroscopy (FT-IR). The chemical structure of unprocessed, acid-treated, PFA-treated,



bleached, and CNC extracted from RH was investigated using FT-IR. Samples were placed on KBr discs, and a Nicolet 6700 spectrometer (M/S Thermo Fisher Scientific Instruments, USA) was used to record spectral signals with an average of 32 scans, 4 cm⁻¹ resolution. OMNIC 9.2.86 (Thermo Fisher Scientific, Inc., USA) software was used to smooth the obtained spectra.

2.3.2. X-ray diffraction (XRD). XRD was used to reveal the structural modifications of materials at every stage of treatment. XRD (D2 PHASER, Bruker, Germany) spectra with Cu-K α (λ = 0.15418 nm), 30 kV, and 10 mA were utilized to calculate the crystal size and crystallinity fraction. The dried samples were put on the quartz base, and scans were taken at a speed of 0.02° min⁻¹ over a 2θ range of 10°–80°. Duplicate XRD analyses were performed on each material. The XRD diffractograms were smoothed and analyzed with Xpert HighScore Plus (PANalytical Technologies Pvt. Ltd, Netherlands) software. The Gauss function in OriginPro 9.0.0 (64 bit) (OriginLab Corporation, USA) was used to fit the curve. The percentage of crystallinity (CrI) was calculated using eqn (1) below based on the areas beneath the crystalline and amorphous peaks after baseline correction.⁴⁷

$$\text{CrI (\%)} = A_c / (A_c + A_a) \times 100 \quad (1)$$

where A_a and A_c are the areas under the amorphous and the crystalline peaks, respectively.

Scherrer's eqn (2) was applied to calculate the dimension D_{hkl} of the crystallites parallel to the hkl diffracting planes.

$$D_{hkl} = (K \times \lambda) / (\beta_{1/2} \times \cos \theta) \quad (2)$$

where correction factor K is 0.9, λ , θ , and $\beta_{1/2}$ are the wavelength, the diffraction angle, and the peak width at half maximum intensity, respectively.

2.3.3. Thermal analysis (TGA). TGA was used to compare the decomposition characteristics of materials at different stages of treatment. The temperature stability of each sample was measured with a TGA/DTA thermogravimetric analyzer (Mettler Toledo Corporation, Schwarzenbach, Switzerland), and the results were analyzed with STAR software (Version 9.3). Runs were done by putting between 2 and 4 mg of the sample into a 70 μ L aluminum cup. Just before the analysis, the cup was crimped, and the instrument automatically poked a hole in it. The investigation was conducted with a steady nitrogen supply of 40 mL min⁻¹ to ensure that the weight loss was caused by thermal degradation. The temperature of the sample was ramped at a constant rate of 10 °C min⁻¹, and weight loss (TGA) or heat evolved (DTA) was measured against increasing temperature.

2.3.4. Transmission electron microscopy (TEM). TEM (JEM-1400 F, JEOL Ltd, Tokyo, Japan) observations were carried out in duplicate at 100 kV accelerating voltage to study the morphology of CNC. Micropipette drops of 1.0 wt% CNC suspension were placed on copper grids, dried at ambient temperature, and examined using TEM.

2.3.5. Raman spectroscopy analysis. The Raman spectra were taken with laser Raman microspectroscopy (XploRA Plus, Horiba) at an excited wavelength of 785 nm and a power of 0.5

mW to prevent the laser heating from affecting the Raman shift of the CNC material. The spectrum was obtained by aggregating three separate scans of distinct CNC sample regions. All spectral changes were done with Origin Pro 9.0.0 (OriginLab, USA).

2.3.6. Solid-state ¹³C nuclear magnetic resonance (¹³C-NMR). Solid-state ¹³C-NMR experiments were performed on a Bruker AVANCE 400 spectrometer using cross-polarization, high-power proton decoupling, and magic angle spinning (CP/MAS). ¹³C-NMR spectra were obtained at 298 K using a 4 mm probe at 100.13 MHz. The CP contact periods were 2 ms, the repetition duration was 1 s, and the MAS rotation frequency was 12 kHz.

3. Results and discussion

3.1. Physical appearance of RH at different treatment stages

According to Fig. 1, the color of RH changed from light brown to reddish brown following treatment with acid and PFA and then to a clear white following bleaching because of the elimination of hemicellulose, lignin, and other non-cellulosic substances. RH was subjected to an acidic environment to partially dissolve hemicellulose, wax, and pectin. Rest and lignin swelled in the acidic environment and migrated toward the surface, facilitating the PFA process. The sample color at this stage was a deeper reddish brown than the original because the phenol derivatives found in lignin absorb visible light.

The PFA phase softened the fiber by removing a significant portion of hemicellulose and some lignin due to oxidant H₂O₂. Most RH hemicellulose, wax, and pectin have been extracted using acid and PFA processing. However, unlike when the fiber was treated with HCOOH, the lignin stayed outside of it, swelled, and migrated to the exterior of the fiber, giving the fiber a deeper color. Therefore, further bleaching is required to eliminate lignin and obtain purified cellulose. Cellulose fiber is a semi-crystalline material made up of solid and amorphous parts.

Consequently, the final step involves acid hydrolysis, in which the acid targets prioritize attacking amorphous regions while avoiding crystalline areas. The primary function of acids is to liberate hydronium ions (H⁺) to degrade the glycoside and ether bonds in the cellulose molecular chain of the amorphous region. After acid hydrolysis, colloidal suspensions of CNC were homogeneous and stable.

3.2. FTIR

The chemical composition of materials undergoing various chemical treatments has been directly determined using FTIR spectroscopy. The FTIR spectra of raw, acid-treated, PFA-treated, bleached, and acid hydrolysis RH are shown in Fig. 2. The prominent bands found in the FTIR spectra of the various treated RH samples are thoroughly summarized in Table S1.† The peak in the 3440–3450 cm⁻¹ range attributed to O–H stretching vibrations of cellulose molecules was seen in all samples. This observation emphasizes the fibers' underlying hydrophilic nature. The C–H bond's asymmetric stretching vibration has a designated frequency range of 2900–



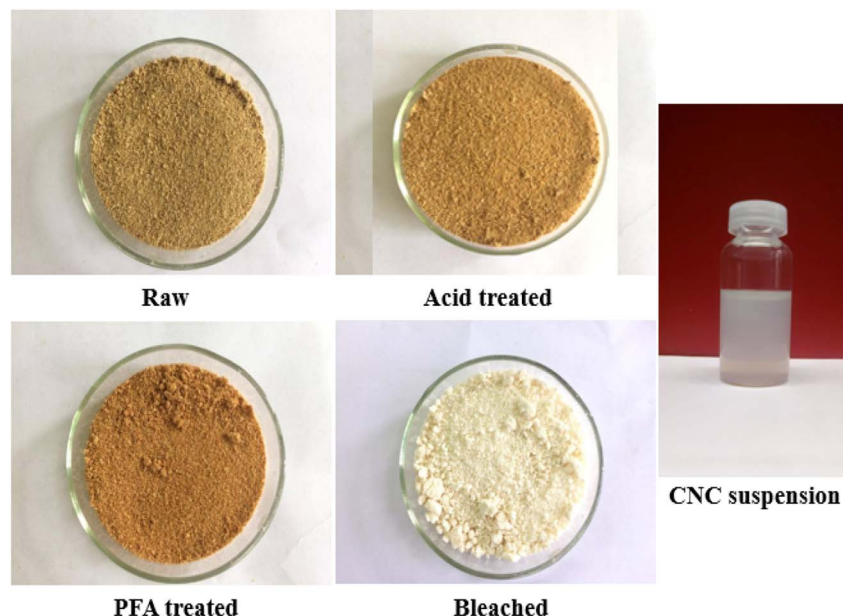


Fig. 1 RH photographs at various phases of chemical treatment and CNC suspension.

2937 cm^{-1} .⁴⁸ A shoulder at 1740 cm^{-1} in RH spectra revealed acetyl and ester in hemicellulose or carboxylic acid groups in ferulic and *p*-coumaric lignin.^{49–51}

The hemicellulose components of RH were dissolved when exposed to HCOOH and PFA acid at 80 °C. The lignin component became flatter and moved closer to the surface. This made the 1740 cm^{-1} peak in the PFA-treated sample more noticeable than in the raw RH sample. This peak disappeared from the spectra of the bleached and acid hydrolysis samples, indicating that the hemicelluloses and lignin in the fibers were extracted

chemically. The perceptible alterations in the location and shape of the FTIR peaks demonstrated the observable structural alteration.

The peak found at 1645 cm^{-1} in all spectra can be assigned to the bending vibration of O–H in the adsorbed water on the cellulose structure.^{52,53} The typical C–O stretching vibration of lignin and hemicellulose of raw RH was found at 1247 cm^{-1} ^{51,53} and was absent from the bleached and acid-hydrolyzed samples, indicating the sufficient removal of hemicellulose lignin components during the chemical treatments. The

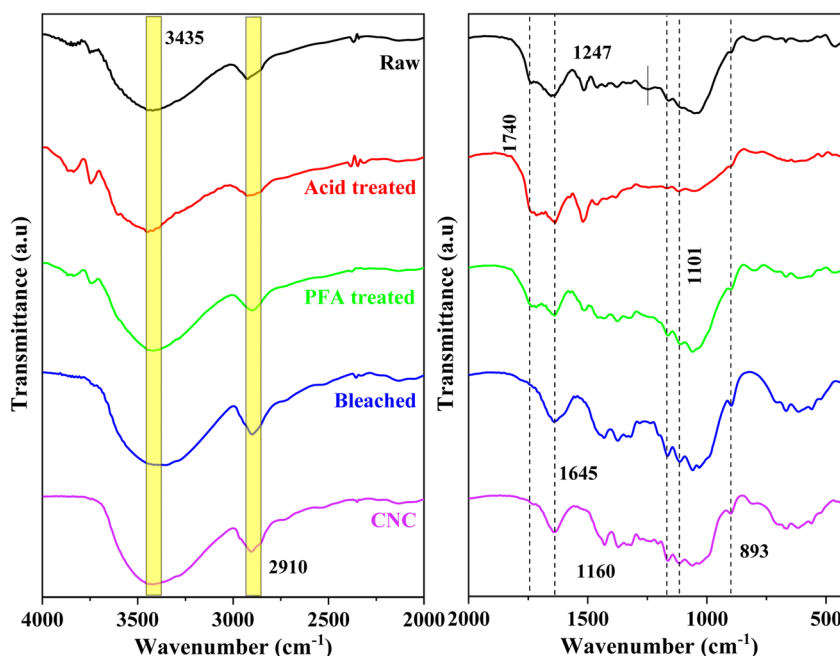


Fig. 2 FTIR spectra of RH at distinct chemical treatment stages and CNC.

deformation vibration of the C–C bond in the ring and the glycoside ether bond (C–O–C) in the β -1,4-glycosidic bond of cellulose were assigned to at 1160 cm^{-1} and 1101 cm^{-1} .⁵⁴ The absorption recorded in all $1024\text{--}1047\text{ cm}^{-1}$ spectra corresponded to the cellulose component's C–H stretching vibration.⁵² The results revealed that the cellulose molecular structure remained unchanged following acid hydrolysis. The spectra's signal at 893 cm^{-1} was linked to the cellulose glucose ring's glycosidic linkages.⁵⁵ This peak was more noticeable after bleaching and acid hydrolysis, indicating that the cellulose content had grown due to further extraction steps. The spectral values discovered agreed with the available literature.⁵⁵ The FTIR spectra of the bleached and acid hydrolysis samples without contaminants were compared to those of the unprocessed RH based on the discussion and analysis that came before them. For example, the RH peaks at 1740 cm^{-1} and 1247 cm^{-1} were no longer present. According to the findings, lignin and hemicellulose were eliminated using chemical treatment.⁵⁶

3.3. XRD

XRD patterns can be utilized to examine the crystalline structure of the untreated and chemically treated cellulose fibers, as illustrated in Fig. 3. All diffractograms showed two peaks, at around 22° and 15° , that corresponded to the crystalline structure of cellulose I.⁵⁷ Hydroxyl groups in cellulose can form intermolecular and intramolecular bonds, leading to cellulose crystalline structures.⁵⁸ This bonding prevents the movement of cellulose chains and makes them align tightly and orderly, leading to a propensity for crystallinity. In unprocessed fiber, these crystalline regions are surrounded by amorphous parts like hemicelluloses, lignin, and pectin, making the fiber less crystalline. The pattern indicated that the crystallinity improved due to the effective removal of non-crystalline components using a series of chemical treatments.

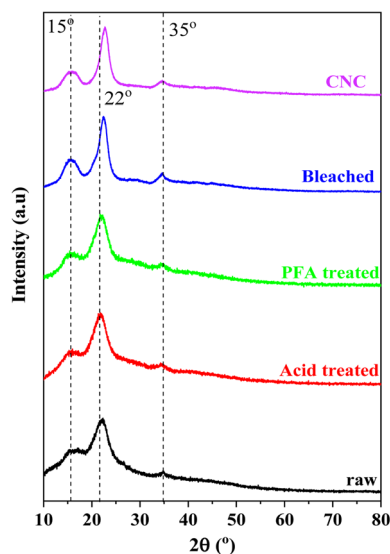


Fig. 3 XRD pattern of RH at distinct treatment stages.

Table 1 CrI and crystalline domain size of RH at different stages of treatment

RH stage	CrI (%)	Crystalline domain size (nm)
Raw	22.63	2.64
Acid-treated	32.04	2.76
PFA-treated	39.33	2.88
Bleached	48.63	3.92
CNC	67.16	3.72

The most conservative XRD phase analysis method for determining cellulose crystallinity was peak deconvolution.^{59–61} As illustrated in Fig. S1,[†] the peak deconvolution technique was used to isolate specific crystalline peaks from the XRD curves to determine the crystallinity of the materials manufactured at various phases using the XRD spectra. Gaussian, Lorentzian, and Voigt functions were typically utilized in deconvolution.⁶¹ All peaks were Gaussian fit in this investigation, and the broad peaks at about 20 and 40° were assigned to the amorphous contribution. In cellulose, an amorphous peak was frequently detected in the 2θ range of $16\text{--}20^\circ$.^{60–62} This study attributed a broad peak around 35° to the crystallinity component to match experimental data. According to Table 1, the RH treated with formic/peroxyformic acid has increased CrI and crystalline domain size.

Acid and PFA treatment disintegrated the amorphous hemicelluloses and lignin, dramatically raising CrI from 22.63% to 39.33%. Bleaching eliminated the remaining amorphous components, producing a maximum CrI of 48.63%. With their removal, a considerable number of crystalline domains can be realigned. The remaining undeveloped areas were sensitive to acid assault and had random orientations. Hydroxyl ions (H_3O^+) can penetrate amorphous regions and encourage the hydrolytic breakage of cellulose's glycosidic linkages, which releases individual crystallites.^{61,63–65} A prominent and strong peak was seen in the XRD pattern after acid hydrolysis, supporting the significant 67.16% improvement in crystallinity. The cellulose isolation methods, the hydrolysis process length, the raw materials used in its synthesis, and the data assessment strategy significantly impacted CNC's crystallinity values.⁶⁶ The high crystallinity of CNC was crucial in defining the barrier and mechanical (strength, stiffness, and rigidity) qualities of biopolymer composites utilized as fillers in industrial applications.^{64,67}

3.4. TGA analysis

Thermogravimetric analysis was performed on untreated RH, acid-treated, PFA-treated, bleached, and CNC samples to examine the degradation characteristics at various preparation phases. Fig. 4a depicts the TG curves for sample weight loss at 25 to 800°C temperatures. Due to the presence of non-cellulosic and cellulosic molecules with different degradation, the thermal decomposition of unprocessed RH was a multi-stage process. The corresponding data for samples in each degradation step are summarized in Table 2. Initial weight loss was



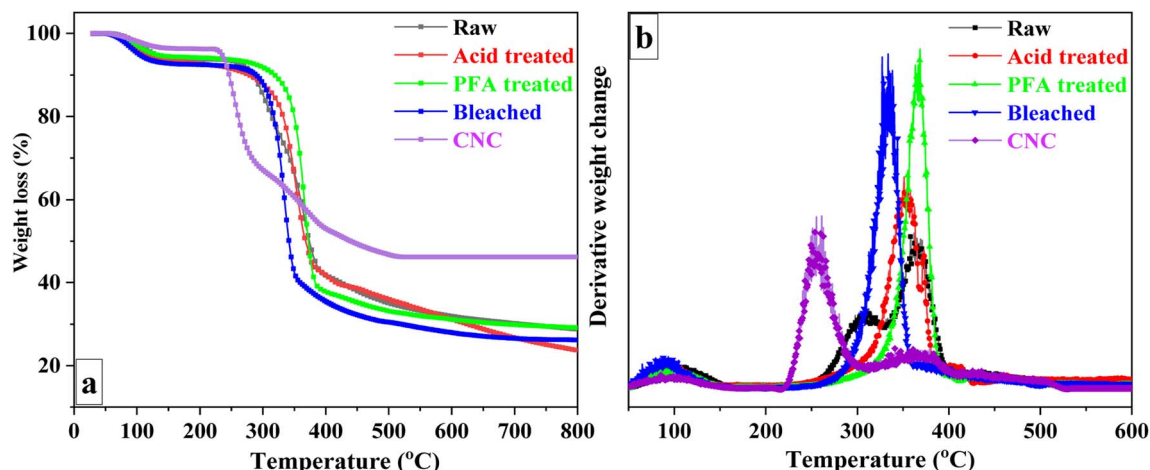


Fig. 4 TGA (a) and DTG (b) curves of RH at different treatment phases.

caused by the evaporation of adsorbed moisture on the surface of RH and intermolecular hydrogen-bonded chemisorbed water between 70 and 120 °C.^{68,69} Thermal depolymerization of hemicelluloses and the breaking of glycosidic linkages in cellulose started the next degradation step at 271 °C.⁵⁸

Lignin is more heat stable than cellulose and hemicelluloses due to its aromatic backbone. Thus, lignin, the primary cementing component, was connected to the third breakdown stage. Raw RH's DTG (Fig. 4b) displayed two broad peaks (at 308 °C and 364 °C), represented the various thermal stabilities and fracture temperatures corresponding to the oxygen functional groups in the lignin structure,⁷⁰ and a low-intensity peak at 425 °C. The acid-treated sample began to degrade at 310 °C and peaked at 352 °C. The PFA-treated sample, on the other hand, began at a higher temperature and reached its highest point at 366 °C. The increase in degradation start temperature was related to flame-retardant lignin on the surface. The bleached RH DTG curve (Fig. 4b) demonstrated a significant weight loss in the range of 280 and 360 °C, with the maximum

degradation peak due to the degradation of cellulose being observed at 334 °C.

In contrast to bleached RH, CNC exhibited a significant difference in decomposition behavior. It was discovered that degradation occurred at a lower temperature than bleached RH. The reduced thermal stability can be attributed to two factors: (1) the substantial surface area of the nanocellulose crystals, which increases the surface area exposed to heat,^{67,71,72} and (2) the presence of sulfate groups on the CNC surface, which exert a catalytic influence and decrease the activation energy required for cellulose chain degradation.⁷³ Furthermore, the crystalline structure of CNC⁷⁴ and the flame-retardant properties of sulfate groups⁷⁵ may contribute to forming more char residue in bleached RH.

The thermal behaviors of samples when heated between 25 °C and 800 °C are shown by the DTA curve in Fig. 5. The endothermal peak at roughly 100 °C in the raw RH indicated that adsorption moisture was evaporating. In addition, the degradation of RH's hemicellulose, cellulose, and lignin components

Table 2 Onset temperature (T_{onset}), weight loss (W_L), degradation temperature at max weight loss (T_{max}), and char yield for RH at different treatment stages evaluated from TG

		Raw	Acid treated	PFA treated	Bleached	CNC
Step I	T_o (°C)	77	70	72	66	70
	T_{max} (°C)	108	98	98	90	98
	W_L (%)	6.3	6.6	5.6	7.4	3.7
Step II	T_o (°C)	271	—	—	—	236
	T_{max} (°C)	308	—	—	—	255
	W_L (%)	18.9	—	—	—	30.3
Step III	T_o (°C)	348	310	327	310	280
	T_{max} (°C)	364	352	366	334	371
	W_L (%)	33.5	49.5	56.5	59.4	13.8
Step IV	T_o (°C)	413	385	384	—	390
	T_{max} (°C)	425	402	430	—	428
	W_L (%)	7.1	3.8	4.2	—	5.8
Char yield (%) at 800 °C		28.7	23.6	29.2	26.2	46.2

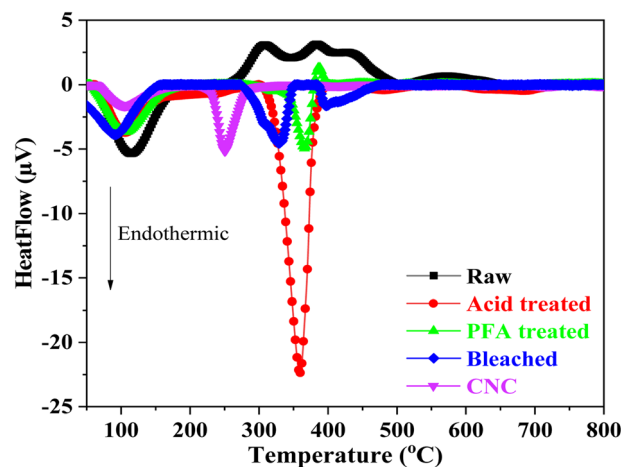


Fig. 5 DTA curves of RH at different stages of treatment.

was linked to two prominent exothermal peaks that overlapped between 250 °C and 500 °C.

The endothermal peak of the acid and PFA-treated samples' DTA curves below 150 °C suggested that the samples' adsorbed moisture had evaporated, which was in line with the weight loss seen in the TGA analysis. TGA and DTA investigations showed that two samples had less adsorbed water than raw RH. When the hemicellulose components of RH were subjected to HCOOH acid and HCOOH/H₂O₂ acid at 80 °C, they disintegrated. The lignin component was flattened and moved toward the surface, suppressing moisture adsorption capacity due to the hydrophobic property of lignin.

Based on the DTA results, it can be concluded that the formic/peroxyformic acid treatment RH underwent endothermal thermal degradation that occurred at higher temperatures than untreated RH. After the PFA treatment, the RH was bleached in a NaOH and H₂O₂ solution. The breakdown of H₂O₂ into H⁺ and perhydroxyl ions (HO₂[−]) was accelerated by the presence of NaOH. These ions attacked the double bonds on the chain and aromatic ring derivatives in lignin, breaking the outer lignin structure and releasing the cellulose component inside the RH.

The DTA analysis of the bleached thermogram revealed an endotherm below 150 °C, which can be ascribed to water loss. A second endothermic transition, which takes place between 280 °C and 360 °C, signifies the fusion of crystallites and reflects the characteristic decomposition of cellulose within this temperature range.³⁸ DTA investigations also revealed a significant difference in thermal behavior between CNC and bleached RH. A broad endothermic range of 280 °C to 360 °C was observed for the bleached sample. In contrast, the CNC exhibited an abrupt and narrow endotherm beginning at 230 °C with a peak at 250 °C. This was probably due to the replacement of hydroxyl groups by sulfate groups (O–SO₃H) during the hydrolysis step. The literature has documented that the sulfate groups found in CNC facilitate direct solid-to-gas phase transitions in nanocellulose crystals.^{76,77} This leads to a gradual decomposition process preceding the pyrolysis of the primary cellulose. The crystalline structure and particle size are drastically altered during sulfation, resulting in an earlier fusion

initiation in CNC than bleached RH. These findings were relevant to the TGA analysis.

3.5. Transmission electron microscopy analysis

TEM observations of the morphology of bleached RH and CNC are depicted in Fig. 6. It validated the data showing that RH-isolated cellulose nanofibrils existed. Long fibrils with widths in the nanometer range (1–10 nm) were portrayed in the photograph. The length-to-diameter or aspect ratio substantially impacts how well nanofibers can reinforce materials. Aspect ratios for most fibers ranged from 120 to 140. This value was relatively high compared to cellulose nanofibrils isolated from other sources, such as wheat straw (90–110)⁷⁸ or coconut shell (60).⁷⁹ As shown in Fig. 6b, acid treatment successfully hydrolyzed and removed the amorphous region in cellulose nanofibrils, causing a substantial length reduction and hydrolyzed cellulose-generated rod-like structures. CNC, generated from RH, had an aspect ratio of 10.2 ± 6.8 , a diameter of 19 ± 3.3 nm, and a length of 195 ± 24 nm. The aspect ratio determines how well nanofibers can act as reinforcement. CNC's more excellent aspect ratio may be employed as a filler for reinforcement within the film to strengthen the biocomposite's mechanical properties.⁸⁰ Similar aspect ratios were reported for CNC extracted from grape skin (5–10), rice straw (8.8–10.5),⁸¹ Mengkuang leaves (10–20),⁸² raw cotton linter (19),⁸³ degreasing cotton (12 ± 10), and waste cotton cloth (17 ± 15).⁸⁴ Roohani *et al.* isolated a CNC with an average width, length, and aspect ratio of 14.6 nm, 171 nm, and 11–12.⁸⁵ According to Chandra *et al.*, cellulose nanofibers made from areca nut husk fibers exhibited a high aspect ratio of 120–150 and a 1–10 nm diameter.⁸⁶

In the current research, the yield of CNC extracted from RH ranged between 28% and 30%, more significant than the yield (12%) of CNC extracted from RH in a previous study.⁸⁷ The utilization of various acids and hydrolysis conditions may be the cause of the yield difference. This yield was similar to the CNC obtained from okra fibers (30%) and Mengkuang leaves (28%).^{82,88} In contrast, the percentage yield of CNC derived from RH was lower than that of CNC obtained from areca nut fibers

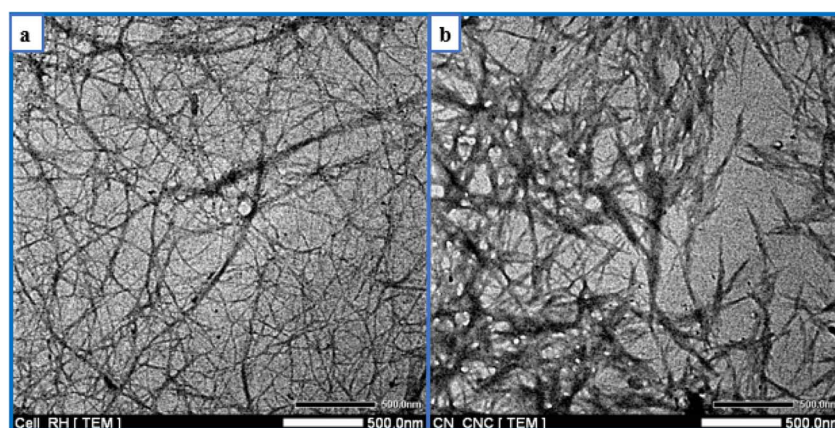


Fig. 6 TEM images of (a) bleached RH and (b) CNC.



($35 \pm 0.2\%$)⁸⁹ and sugarcane bagasse (66.45%)⁹⁰ and higher than that of CNC obtained from barley straw and husk (12%)⁹¹ and old corrugated container fiber (23.98%).⁹² The cellulose source and the isolation conditions have the most significant impact on CNC yield. It is possible that the use of mechanical pre-treatment on cellulose, combined with acid hydrolysis under precisely controlled conditions, including acid concentration, hydrolysis time, and temperature, and followed by post-treatment procedures, such as homogenization and ultrasonication, will increase the production yield of CNC.⁹³

3.6. Raman analysis

Due to its high sensitivity to changes in the hydrogen bonding environment, Raman spectroscopy was used to detect changes in the polymorphic arrangement of cellulose.⁹⁴ The Raman spectra obtained from three separate scans carried out on various areas of the CNC sample are shown in Fig. 7a. The resulting band pattern resembles those found in earlier research.^{94,95} Bands produced by the vibrations of COC, CH₂, CH, and OH groups, as well as the likely vibrations of CH-O and OH-O hydrogen bonds, predominated in this range between 250 and 1600 cm⁻¹. The bands that best reflected the molecular surroundings of the methylene (-CH₂) and methine (-CH) groups were 1478–1464 cm⁻¹, 1376–1337 cm⁻¹, 1263–1286 cm⁻¹, and 980 cm⁻¹. CH₂ scissoring (bending) deformations and OH bending deformations at C-6 were principally responsible for bands at 1478 cm⁻¹ and 1464 cm⁻¹.^{96,97} These bands revealed details on cellulose's intramolecular and intermolecular bonds' ultrastructure.⁹⁴

The glycosidic linkage orientation was sensitive between 1200 and 1300 cm⁻¹. A weak band at 1286 cm⁻¹ and a strong band at 1263 cm⁻¹ were discernible in this area (Fig. 7c). While the latter was connected to cellulose II, the former has been

seen before in bacterial and wood cellulose I. The CH₂ group vibrations were linked to the bands at 980 cm⁻¹. C-O-C glycosidic linkage deformations caused the 1121, 1092, and 490 cm⁻¹ bands. Ring deformations were apparent between 541 and 1147 cm⁻¹ (Fig. 7b and c). The 435 and 421 cm⁻¹ bands were connected to CCC and CCO ring vibrations, respectively. Bands at about 435 cm⁻¹ were typical of cellulose I, whereas bands at 421 cm⁻¹ were unique to cellulose II.⁹⁸

Agarwal proved that bands at about 380 cm⁻¹ are related to the cellulose crystallinity index.⁹⁵ Bands at 327 cm⁻¹ and 580 cm⁻¹, respectively, are other skeletal deformations indicative of cellulose I (Fig. 7b). The bending vibration of the methine group at C-1 or the bending vibration of HCC and HCO at C-6 in the pyranose unit were responsible for the broadband between 840 and 880 cm⁻¹.⁹⁸

3.7. ¹³C-NMR analysis

Carbon from cellulose has ¹³C NMR signals between 60 and 110 ppm. Fig. 8 depicts the CNC spectrum extracted from RH. The peaks at 72.4 ppm and 73.6 ppm were cellulose carbons C-2 and C-3, respectively. The peak at 75.5 ppm was cellulose carbon C-5. The peak represented crystalline cellulose carbon C-4 at 89.3 ppm, while the peak represented crystalline cellulose carbon C-1 at 105.5 ppm.⁹⁹

The C-6 peak suggested crystalline cellulose at 65.7 ppm.¹⁰⁰ The NMR spectra showed that most hemicellulose and lignin were removed after chemical processes (acid treatment, PFA treatment, bleaching, and hydrolysis). Eliminating C-1 from the hemicellulose's (1-4)-D-Xylp-2-O-(4-OMe-dGlcP) units cause the 102 ppm peak to be absent.¹⁰¹ Furthermore, the absence of representative peaks of hemicellulose and lignin at 21, 56, or 173 ppm¹⁰² suggested that the chemical treatments removed the carbons in lignin's methyl, methoxyl, and carboxylic groups.

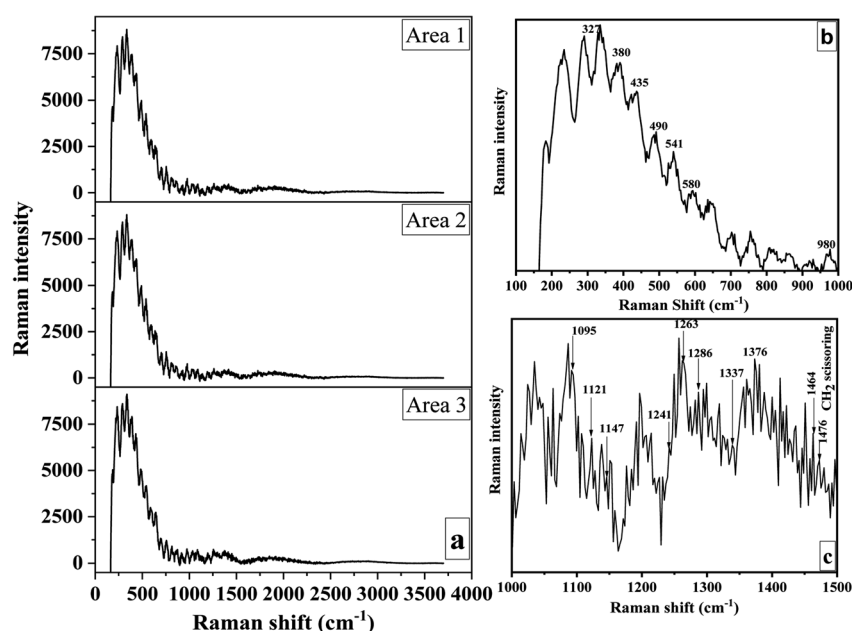


Fig. 7 Raman spectra of three separate scans of distinct CNC regions (a) and within specified band regions (b and c).

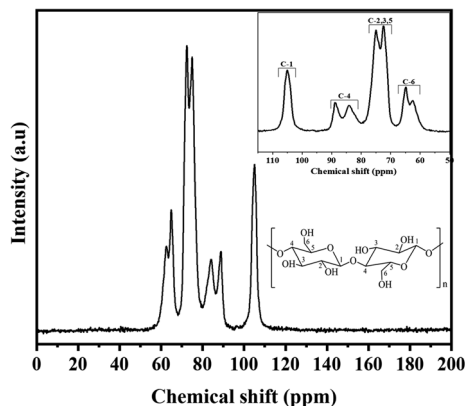


Fig. 8 ^{13}C NMR spectra of CNC.

In conclusion, the spectra of CNC elucidated that each peak originated from six carbon atoms within the cellulose structure.

4. Conclusion

Using a formic/peroxyformic procedure, we successfully extracted cellulose from rice husk (RH). The technique above successfully eliminated non-cellulose components while also inducing swelling and migration of lignin toward the RH's surface, minimizing the requirement for recurrent bleaching and shortening the bleaching process. To create cellulose nanocrystals (CNC), 64% sulfuric acid was used to hydrolyze the cellulose. With the CNC, cellulose's crystallinity rose from 22.63% with the raw RH to 67.16%. The leading cause was the isolation and hydrolysis processes, which eliminated amorphous components like hemicellulose, lignin, pectin, and the fiber structure's loosely organized underdeveloped cellulose region. After H_2SO_4 hydrolysis, a rod-like CNC was produced. With an aspect ratio of 10.2 ± 6.8 , CNC had a diameter and length of 19 ± 3.3 nm and 195 ± 24 nm. According to the research, CNC offers excellent promise for using nanocomposite materials as a reinforcing phase.

Data availability

Data will be made available on request.

Author contributions

An Nang Vu: conceptualization, investigation, writing – original draft preparation. Long Hoang Nguyen: formal analysis. Ha-Chi V. Tran: formal analysis. Kimio Yoshimura: resources. Tap Duy Tran: investigation. Hieu Van Le: writing – review & editing. Ngoc-Uyen T. Nguyen: conceptualization, validation, supervision.

Conflicts of Interest

The authors report no declarations of interest.

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